

Answer Key — Biology Grade 11

Sub-Strand 2.1: Reproduction in Plants • Lesson 7: Putting the Evidence Together: Why Cassava Cuttings Need No Seeds but Lone Pawpaw and Avocado Trees Need a Partner

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	B	C	D	D	A	B	C

USE AFTER: PREDICT PHASE — Predict: why does a cassava cutting not need flower parts?

Q1. A farmer in Kisumu plants a piece of cassava stem in the soil. It grows into a whole new plant without any flower, pollen or seed. Which statement best explains this?

- A. The stem cutting must first be pollinated by insects before it can grow
- B. The cutting grows only because it has a flower that is too small to see
- C. New roots and shoots grow from the stem tissue of the parent, so no flowers or fertilisation are needed**
- D. The cutting is really a seed that is hidden inside the stem

Correct answer: C. Vegetative propagation (asexual reproduction) grows a new plant from a part of the parent, so no gametes, pollination or fertilisation are needed. A cutting is not a seed.

USE AFTER: OBSERVE PHASE — Observe: hibiscus flower dissection, features that hinder self-pollination

Q2. In a hibiscus flower, the anthers release their pollen before the stigma of the same flower is ready to receive it. What is this feature called, and what does it do?

- A. Dioecy: male and female flowers are on separate plants
- B. Protandry: the timing difference makes self-pollination less likely**
- C. Monoecy: male and female flowers are on the same plant
- D. Herkogamy: the stigma and anthers are at different heights

Correct answer: B. Protandry means the anthers mature and release pollen before the stigma is receptive. This hinders self-pollination. Herkogamy is a difference in height, not in timing.

USE AFTER: OBSERVE PHASE — Observe: labelled chart comparing pawpaw (dioecious) and maize (monoecious)

Q3. A maize plant has male tassels at the top and female cobs with silk lower down on the same plant. Which description is correct?

- A. It is protandrous, because the cob grows from the tassel
- B. It is self-incompatible, because it has no male parts
- C. It is monoecious, because separate male and female flowers grow on the same plant**
- D. It is dioecious, because it has male and female parts on separate plants

Correct answer: C. Monoecious plants such as maize have separate male flowers (tassels) and female flowers (cobs) on the same plant. Pollen moves by wind from the tassel to the silk. Dioecious plants have them on separate plants.

USE AFTER: OBSERVE PHASE — Observe: pre-downloaded animation of double fertilisation, paused at pollen tube growth

Q4. During double fertilisation in a flowering plant, two sperm nuclei from the pollen tube enter the ovule. What does each one do?

- A. One forms the seed coat and the other forms the fruit wall
- B. Both fuse with the egg to form two embryos
- C. One fuses with the egg to form the endosperm, and the other fuses with the polar nuclei to form the zygote
- D. One fuses with the egg to form the zygote, and the other fuses with the polar nuclei to form the endosperm**

Correct answer: D. One sperm nucleus fuses with the egg to form the zygote (the future embryo). The other fuses with the polar nuclei to form the endosperm, the food store in seeds such as maize.

USE AFTER: EXPLAIN PHASE — Explain: Think-Pair-Share on placentation of tomato, passion fruit and bean pod

Q5. A bean pod is opened and the seeds are found attached in a row along one edge of the pod. What type of placentation is this?

- A. Parietal placentation
- B. Free-central placentation
- C. Axile placentation
- D. Marginal placentation**

Correct answer: D. In beans, ovules attach along one edge (suture) of the single-chambered ovary, which is marginal placentation. Axile is seen in tomato, and parietal in passion fruit.

USE AFTER: EXPLAIN PHASE — Explain: CER statement for classifying a fruit

Q6. After pollination and fertilisation in a tomato flower, which part of the flower develops into the fruit and which part develops into the seeds?

- A. The ovary becomes the fruit and the ovules become the seeds**
- B. The ovules become the fruit and the ovary becomes the seeds
- C. The petals become the fruit and the anthers become the seeds
- D. The stigma becomes the fruit and the style becomes the seeds

Correct answer: A. The fertilised ovary develops into the fruit, and the fertilised ovules inside it develop into seeds. This is why fruit forms only after pollination and fertilisation.

USE AFTER: EXPLAIN PHASE — Explain: comparing advantages and disadvantages of cuttings and seeds

Q7. A farmer in Kericho plants a whole field with cassava cuttings taken from one parent plant. A disease that attacks this variety arrives. Why is the whole field at high risk?

- A. Cuttings produce seeds that carry the disease
- B. Cuttings from one parent are genetically identical, so they all lack resistance to the disease**
- C. Cuttings need pollinators, and the disease kills them
- D. Cuttings always grow more slowly than seeds, so they are weak

Correct answer: B. Asexual reproduction produces offspring identical to the parent with no variation, so a disease can spread through all of them. Sexual reproduction gives variation, so some plants may resist the disease.

USE AFTER: DQB PHASE — DQB: posting an evidence-based answer to why the lone pawpaw or avocado has flowers but no fruit

Q8. A lone avocado tree in a compound in Kiambu is covered in flowers every year but never sets fruit. Which explanation best uses what we know about barriers to self-pollination?

- A. Flowers are only for decoration, so they never lead to fruit
- B. Avocado trees can only be grown from cuttings, so they cannot form fruit
- C. Its flowers cannot be fertilised by its own pollen (self-incompatibility), so it needs pollen from another avocado tree**
- D. The tree has too many petals, which stop the ovary from growing

Correct answer: C. Barriers such as self-incompatibility and flowers maturing at different times stop a lone tree from fertilising itself. Pollen from a second tree is needed before the ovary can develop into fruit.

